INL1, Problem 2

MAA155

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(a) Two examples of ideals in $\mathbb{Z}$:

we know, from previous knowledge of ideals, that any ideal I in $\mathbb{Z}$ can be written as $I = n\mathbb{Z}, n \in \mathbb{N}$

so let's take for example $I = 3\mathbb{Z}$ and $J = 5\mathbb{Z}$

(it's clear that they are ideals since they are groups and $3k \cdot r = 3k' \in 3\mathbb{Z}$, $5k \cdot r = 5k'' \in 5\mathbb{Z}$)

Also, $3 \in 3\mathbb{Z}$ and $5 \in 5\mathbb{Z}$ but $3 \notin 3\mathbb{Z} \cap 5\mathbb{Z}$ and $5 \notin 3\mathbb{Z} \cap 5\mathbb{Z}$

so $I \cap J \neq I$ and $I \cap J \neq J$

(b) Let I and J be ideals of R

* $I \neq \emptyset$ then $\exists i \in I$, for the same reason $\exists j \in J$

using the ideal "absorption" propriety : $i.j \in I$ and $i.j \in J$ then $i.j \in I \cap J$ so $I \cap J \neq \emptyset$

* let $a, b \in I \cap J$, so $a \in I$ and $b \in I$, then $a+b \in I$ since I is an ideal, for the same reason $a+b \in J$,
so $a+b \in I \cap J$

* let $r \in R$, $ar \in I$ (I ideal), but $ar \in J$ (J ideal) so $ar \in I \cap J$

$I \cap J$ is an ideal of R ▨

(c) let $a, b, c \in R$

$$\left.\begin{array}{l} a \mid b \Rightarrow \exists k \in R,\ b = ka \\ b \mid c \Rightarrow \exists k' \in R,\ c = k'b \end{array}\right\} \Rightarrow \exists k, k' \in R,\ c = k'ka \Rightarrow \exists K \in R,\ c = Ka \Rightarrow a \mid c$$

▨